Lab Sheet 1 — Basic Linux Commands — File and Directory Operations

Complete questions with answers immediately below each question. Commands are written for common Linux distributions; use sudo only when required.

Lab Sheet 1 — Basic Linux Commands — File and Directory

Operations

distributions; use sudo only when required.

Lab Sheet 1 — Basic Linux Commands — File and Directory

Operations

sudo only when required.

Question 1: Print your current working directory to verify your location in the filesystem.

Answer / Solution:

```
pwd
```

Question 2: List all files and directories in your current location using the basic ls command.

Answer / Solution:

```
ls
```

Question 3: List all contents of your home directory, including hidden files (starting with a dot), using appropriate flags.

Answer / Solution:

```
ls -la ~
```

Question 4: Display all files in long-listing format with human-readable file sizes using ls.

Answer / Solution:

```
ls -lh
```

Question 5: Create a new directory named myfirstlab in your current directory.

Answer / Solution:

```
mkdir myfirstlab
```

Question 6: Navigate into the newly created myfirstlab directory using cd.

Answer / Solution:

```
cd myfirstlab
```

Question 7: Inside myfirstlab, create three empty text files: a.txt, b.txt, and c.txt using a single command.

Answer / Solution:

```
touch a.txt b.txt c.txt
```

Question 8: Display the full contents of the system file /etc/os-release on your terminal screen.

Answer / Solution:

```
cat /etc/os-release
```

Question 9: Check the username of the user currently logged into the terminal session.

Answer / Solution:

```
whoami
```

Question 10: Display your system's hostname on the terminal.

Answer / Solution:

```
hostname
```

Question 11: Check how much disk space is currently free on your system in human-readable format.

Answer / Solution:

```
df -h
```

Question 12: Check the total system RAM and current memory usage in human-readable units.

Answer / Solution:

```
free -h
```

Question 13: Check how long your Linux system has been continuously running.

Answer / Solution:

```
uptime
```

Question 14: Display full system and kernel operating information using uname.

Answer / Solution:

```
uname -a
```

Question 15: Use the manual pages to look up the documentation for the ls command.

Answer / Solution:

```
man ls (press q to quit)
```

Question 16: Display a short quick-usage summary for the cp command using the --help flag.

Answer / Solution:

```
cp --help
```

Question 17: Use the whatis command to get a one-line description of the mv command.

Answer / Solution:

```
whatis mv (or: mv --help)
```

Question 18: Navigate back to your home directory using the shorthand tilde (~) symbol.

Answer / Solution:

```
cd ~
```

Question 19: Create a nested folder hierarchy project/src/data in a single command using mkdir.

Answer / Solution:

```
mkdir -p project/src/data
```

Question 20: Delete the file c.txt you created inside the myfirstlab directory.

Answer / Solution:

```
cd ~/myfirstlab && rm c.txt
```

Medium Exercises (Q21–Q35)

Question 21: Create a file named students.txt and add 5 student names (one per line) using echo and the append redirection operator (>>).

Answer / Solution:

```
touch students.txt && printf 'Amit\nNeha\nRavi\nZara\nRohan\n' >> students.txt
```

Question 22: Display only the first 2 names from students.txt using the head command.

Answer / Solution:

```
head -n 2 students.txt
```

Question 23: Display only the last 2 names from students.txt using the tail command.

Answer / Solution:

```
tail -n 2 students.txt
```

Question 24: Create a directory named backup and copy students.txt into it without navigating out of your current directory.

Answer / Solution:

```
mkdir -p backup && cp students.txt backup/
```

Question 25: Rename students.txt to class_list.txt using the mv command.

Answer / Solution:

```
mv students.txt class_list.txt
```

Question 26: Use grep to search inside class_list.txt for any name containing a specific letter (e.g., 'a' or 'R').

Answer / Solution:

```
grep -i 'a' class_list.txt
```

Question 27: Display line numbers along with matching text when searching for a string in class_list.txt using grep -n.

Answer / Solution:

```
grep -in 'a' class_list.txt
```

Question 28: Change permissions of class_list.txt so only the owner has read and write rights (no execute), and verify using long listing.

Answer / Solution:

```
chmod 600 class_list.txt && ls -l class_list.txt
```

Question 29: Find all files ending with .txt in your home directory tree using the find command.

Answer / Solution:

```
find ~ -type f -name '*.txt'
```

Question 30: Use wc -l to count the exact number of lines (students) in class_list.txt.

Answer / Solution:

```
wc -l class_list.txt
```

Question 31: Display the first 5 lines of /etc/passwd on the terminal.

Answer / Solution:

```
head -n 5 /etc/passwd
```

Question 32: Display the last 5 lines of /var/log/dpkg.log or /var/log/syslog.

Answer / Solution:

```
tail -n 5 /var/log/dpkg.log 2>/dev/null || tail -n 5 /var/log/syslog
```

Question 33: Copy an entire directory structure (project/) into a new folder named project_backup/ recursively.

Answer / Solution:

```
cp -r project project_backup
```

Question 34: View a large text file page-by-page using less and identify the key used to quit the pager.

Answer / Solution:

```
less /path/to/largefile.txt (press q to quit)
```

Question 35: Move two files (a.txt and b.txt) simultaneously into a target folder named archive/.

Answer / Solution:

```
mkdir -p archive && mv ~/myfirstlab/a.txt ~/myfirstlab/b.txt archive/
```

Hard / Challenge Exercises (Q36–Q50)

Question 36: Combine commands: Create a directory logs, inside it create app.log with 10 lines of dummy text using a loop or repeated echo >>, then view the last 3 lines using tail.

Answer / Solution:

```
mkdir -p logs && for i in {1..10}; do echo "Dummy log line $i" >> logs/app.log; done && tail -n 3 logs/
app.log
```

Question 37: Use find paired with -exec to locate all .log files in your current working path and display their contents in a single command line.

Answer / Solution:

```
find . -type f -name '*.log' -exec cat {} \;
```

Question 38: Recursively search all files in your current directory tree for the word 'error' using grep -r.

Answer / Solution:

```
grep -rn 'error' .
```

Question 39: Create a shell script file test.sh, grant it execute permissions using symbolic chmod syntax (+x), and write a comment inside explaining the permission string rwxr-xr--.

Answer / Solution:

```
printf '#!/bin/bash\n# rwxr-xr-- means owner=rwx, group=r-x, others=r--\necho Hello' > test.sh && chmod
754 test.sh
```

Question 40: Write a single-line command using chaining (&&) to create a folder demo, enter it, and list its contents.

Answer / Solution:

```
mkdir demo && cd demo && ls
```

Question 41: Combine find and wc -l in a single pipeline to count how many total .txt files exist under your home directory tree.

Answer / Solution:

```
find ~ -type f -name '*.txt' | wc -l
```

Question 42: Find all directories (not regular files) inside /var/log using find -type d.

Answer / Solution:

```
find /var/log -type d
```

Question 43: Search for a specific file system configuration entry inside /etc/fstab using grep, ignoring case sensitivity.

Answer / Solution:

```
grep -i 'ext4' /etc/fstab
```

Question 44: Check the disk space consumption of a specific folder in human-readable format using du -sh.

Answer / Solution:

```
du -sh /path/to/folder
```

Question 45: Write a command that displays the 3 largest files in a directory by sorting ls output by file size.

Answer / Solution:

```
ls -lhS /path/to/directory | head -n 4
```

Question 46: Debug / Safety Habit: Explain why running rm -rf * in an unverified path is hazardous, and state the safety verification command habit that prevents accidental deletion.

Answer / Solution:

```
rm -rf * recursively deletes matching entries without confirmation; first run pwd and ls -la, verify the
exact path and use safer
explicit paths.
```

Question 47: Use touch on an existing file to update its access and modification timestamp to the current system time without modifying its contents.

Answer / Solution:

```
touch existing_file.txt (updates access/modification times; does not change contents)
```

Question 48: Create a file containing multiple duplicate lines, then use sort and uniq to display only unique lines from the file.

Answer / Solution:

```
printf 'pear\napple\napple\nbanana\n' > duplicates.txt && sort duplicates.txt | uniq
```

Question 49: Redirect the stdout of a command (e.g., uname -a) to a file named sys_info.txt, then view its content.

Answer / Solution:

```
uname -a > sys_info.txt && cat sys_info.txt
```

Question 50: Find all files in /var/log that were modified within the last 7 days using find -mtime.

Answer / Solution:

```
find /var/log -type f -mtime -7 -print
```